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Startup Application Form

Winter 2022

COMPANY

Company name:

Ambient Systems

Company url, if any:

https://www.ambientsystems.ai/

If you have a demo, what's the url? Demo can be anything that shows us how the product works. Usually that's a video or screen recording.

(Please don't password protect it; just use an obscure url.)

https://youtu.be/sgD-OnC56zw

Describe what your company does in 50 characters or less.

Applied AI for Cyber Physical Syst

What is your company going to make? Please describe your product and what it does or will do.

There is an increase in demand for intelligent machines that can interact with the environment around them. This involves a highly complex data environment while being embedded in an ever-increasing amount of knowledge in the intelligent systems. Customers look towards effective outputs such as automation-enabled planning, scheduling, predictive maintenance, efficiency and effectiveness of decision-making

Our software provides patented AI methodology to overcome challenges faced by Cyber Physical Systems (CPS) particularly in the built environment. Integrated with partner-manufactured sensors and hardware, we apply machine learning techniques to balance different domains of priorities between desired outcomes and equipment

Where do you live now, and where would the company be based after YC?

(List as City A, Country A / City B, Country B.)

Singapore, Singapore/Japan/Hon

CONTACT

Personal of the founder who is filling out this application:

Please enter an that you check often and that you will have access to for a long time.

ivan@ambientsystems.ai

Phone number(s):

65[redacted]

Edit Your Profile Profile Complete

FOUNDERS

Please provide the personal es of the other cofounders in the startup. No need to add yours again. Founders must have at least 10% equity in the company. We will send an email to each founder to fill out additional information about themselves.

of Cofounder

Remove Profile Complete

Add Cofounder

Please enter the url of a 1 minute unlisted (not private) YouTube video introducing the founder(s). This video is an important part of the application. (Follow the Video Guidelines.)

https://youtu.be/dn88dYtKCeg

Please tell us about an interesting project, preferably outside of class or work, that two or more of you created together. Include urls if possible.

How long have the founders known one another and how did you meet? Have any of the founders not met in person?

How many founders are on the team?

(Fill out this number of founder profiles)

2

CATEGORY

Which category best applies to your company?

Analytics

PROGRESS

How far along are you?

We are currently in the pre-revenue stage, focusing on iterating on our MVP with our niche customers and partners. In the last 6 months, we have completed a 6-digit fundraise from National University of Singapore (NUS), launched our minimum viable product (MVP), signed 2 memorandum of understandings (MOU), and filed a patent for our AI solution. We have set out to fully develop our product for commercialisation by end of the year and start monetizing in Singapore and Japan.

How long have each of you been working on this? How much of that has been full-time? Please explain.

We have been incubating under National University of Singapore for over a year, and have gone full-time right after securing the funding. Apart from the CEO and CSO, 2 full-timers and 2 part-timers assisting with the product and business development have joined us since mid-2021.

Are people using your product?

Yes

How many active users or customers do you have? If you have some particularly valuable customers, who are they? If you're building hardware, how many units have you shipped?

Do you have revenue?

No

Anything else you would like us to know regarding your revenue or growth rate?

If you are applying with the same idea as a previous batch, did anything change? If you applied with a different idea, why did you pivot and what did you learn from the last idea?

If you have already participated or committed to participate in an incubator, "accelerator" or "pre-accelerator" program, please tell us about it.

We have successfully graduated from National University of Singapore's GRIP (Graduate Research Innovation Program) and secured funding from them. Through the program, we were able to fine-tune our target segments from general buildings to users with more specialized needs like healthcare facilities and indoor farms. Apart from the startup learnings provided, we were able to launch our pilot tests with our beachhead customers and also formed collaborative partnerships with manufacturing partners.

IDEA

Why did you pick this idea to work on? Do you have domain expertise in this area? How do you know people need what you're making?

Buildings are a cornerstone of modern society. However, they are not being optimally used, leading to energy wastage, , and overall poor utilization. Existing solutions are either too complex, expensive, or only help the user reduce energy bills, which is a good start but not enough. We believe AI will proliferate across the built environment in the upcoming decade, helping to overcome these challenges. We chose to work on this challenge because we see the potential for AI to transform how we interact with the built environment. CEO's background is computer science and engineering, providing a good understanding of building science and energy systems. Our idea has been validated by users in various

What's new about what you're making? What substitutes do people resort to because it doesn't exist yet (or they don't know

about it)?

Our traction has validated our hypothesis that the technologies of today have yet to meet the needs of our modern CPS and there is research potential in the area of prescriptive maintenance within specific use cases. Most CPS particularly in the built environment still operate in a closed loop system which cuts off communication with operational systems, leading to an information gap between desired outcomes and equipment settings. Most customers simply don't have the in-house expertise to integrate and manage systems like the Building Automation Systems (BAS) offered by Siemens, Honeywell, .

Who are your competitors, and who might become competitors? Who do you fear most?

There are about 10-20 vendors providing AI solutions in the built environment space. Most of them provide it as an add-on feature as part of their existing products, such as building management systems (BMS), lighting systems, or sensors. They are fairly new in the space, with market leaders launching as recently as 2018, focusing mostly on energy savings.

What do you understand about your business that other companies in it just don't get?

Ambient understands that energy consumption is just one of the things that users want to optimize. Sometimes this is in direct competition with other goals e.g. in an indoor farm, there is a mismatch between desired outcomes e.g. crop growth and equipment settings. Likewise, in healthcare facilities, building systems are disconnected from priorities such as patient well-being or hospital associated infections (HAI) preventions. We provide a solution that balances these competing priorities using the latest

How do or will you make money? How much could you make?

(We realize you can't know precisely, but give your best estimate.)

B2B subscription model with optional upfront expenses if the user needs hardware (e.g., sensors, controllers) or system integration. The subscription costs are based on the number of data feeds and AI modules that the user needs. We’re aiming to tap into the rapidly growing AI for Buildings Market (CAGR 36%).

How will you get users? If your idea is the type that faces a chicken-and-egg problem in the sense that it won't be attractive to users till it has a lot of users (e.g. a marketplace, a dating site, an ad network), how will you overcome that?

We believe that research development in this domain is still nascent and building operators of our specialized segment recognize that the benefits of collaboration would lead to perceivable improved operational outputs. This has also been validated through our primary market research with our potential customers in the agriculture and healthcare sectors. Each user will be able to use our product immediately, although ML model accuracy will improve over time. Any additional hardware will take around a day to install, allowing the system to fill in any data gaps immediately.

EQUITY

Have you formed ANY legal entity yet?

This may be in the US, in your home country or in another country.

Yes

Please list all legal entities you have and in what state or country each was formed (e.g. Delaware C Corp, Mexican SAPI, Singapore Pvt Ltd, etc.).

This might include subsidiary companies, legal entities you formed in other markets to do business, entities you created before a pivot, or legal entities you set up to employ developers or other service providers.

Ambient Systems Pte. Ltd. (Incorporated in Singapore on July 5, 2021)

Have you issued stock to investors?

No

Have you raised money using crowdfunding?

No

Please describe the breakdown of the equity ownership in percentages among the founders, employees and any other stockholders. If there are multiple founders, be sure to give the equity ownership of each founder.

Have you taken any investment yet?

Yes

List any investments your company has received. Include the name of the investor, the amount invested, the premoney valuation / valuation cap, and the type of security sold (s, safes or stock).

How much money do you spend per month?

$

How much money does your company have in the bank now?

$

How long is your runway?

(e.g. 5 months)

Please provide any other relevant information about the structure or formation of the company.

LEGAL

Are any of the founders covered by noncompetes or intellectual property agreements that overlap with your project? If so, please explain.

Who writes code, or does other technical work on your product? Was any of it done by a non-founder? Please explain.

All code and technical work was done by (founder and CEO).

Have you received any government grants?

Is there anything else we should know about your company?

(Pending lawsuits, cofounders who have left, etc.)

OTHERS

If you had any other ideas you considered applying with, please list them. One may be something we've been waiting for. Often when we fund people it's to do something they list here and not in the main application.

Please tell us something surprising or amusing that one of you has discovered.

(The answer need not be related to your project.)

Koalas have a 'smooth' brain. This means that they lack higher level recognition and understanding that many other animals have. If you gather a bunch of Eucalyptus leaves, which the koalas eat, and put them on a plate in front of the koala, the koala won't know what to do with them; they just sit there and gawk at it. (Also, baby koalas eat their mother's poop)

CURIOUS

What convinced you to apply to Y Combinator? Did someone encourage you to apply?

We have been in the works with Ambient for a considerable time, and with the traction made in the last half of the year has given us the confidence to apply for Y Combinator.

How did you hear about Y Combinator?

Word of mouth in the start-up env

Resubmit Application